

Eigenvectors & Eigenvalues

Let $X = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, and $\vec{v} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

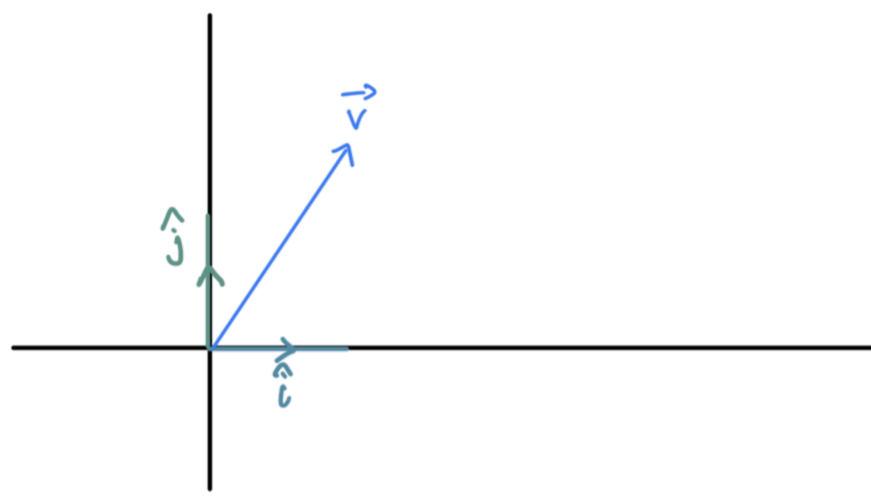
for $X\vec{v} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ -3 \end{bmatrix}$ $X\vec{v} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 0 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

$\vec{v}$ and $\vec{w}$ but $3\vec{v} = \vec{z}$
not linearly related, hence $\vec{v}$ is an eigenvector of X
and 3 is its eigenvalue

form:

$$A\vec{x} = \lambda\vec{x}$$

$\vec{x}$ is the eigenvector of A ,
 λ is the eigenvalue of $\vec{x}$



$$\vec{i} = (1, 0) \quad A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\vec{j} = (0, 1)$$

$$\vec{v} = 2\vec{i} + 3\vec{j}$$

Whenever we multiply a square matrix with a vector,
the resultant is a linear transformation of that vector

$$A \cdot \vec{v} = \text{row } x \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \times 2 + 1 \times 3 \\ 1 \times 2 + 2 \times 3 \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

$A \cdot \vec{v}$ = "rebuild $\vec{v}$ out of the usual basis vectors"

Now, about linearly dependent columns,

$$B = \begin{bmatrix} a & R_a \\ b & R_b \end{bmatrix}$$

$$C = \begin{bmatrix} a & b & a+b \\ p & q & p+q \\ u & w & u+w \end{bmatrix} = \begin{bmatrix} a & b & a \\ p & q & p \\ u & w & u \end{bmatrix} + \begin{bmatrix} a & b & b \\ p & q & q \\ u & w & w \end{bmatrix}$$

$$\det(C) = \det \begin{bmatrix} a & b & a+b \\ p & q & p+q \\ u & w & u+w \end{bmatrix} = \det \begin{bmatrix} a & b & a \\ p & q & p \\ u & w & u \end{bmatrix} + \det \begin{bmatrix} a & b & b \\ p & q & q \\ u & w & w \end{bmatrix}$$

$$\det(C) = 0 + 0 = 0$$

$$A\vec{v} = \lambda\vec{v}$$

$$A\vec{v} - \lambda\vec{v} = 0$$

$$A\vec{v} - \lambda\vec{v} = 0$$

$$(A - \lambda I)\vec{v} = 0$$

$$(M)\vec{v} = 0$$

Vector $\vec{v}$ is dual to
be in machine M 's
null space, on its
kernel

> When looking for a non-zero vector $\vec{v}$ which the
matrix $(A - \lambda I) = M$ transforms to zero vector.

> When can a matrix transform a non-zero vector to zero?

• Only if the matrix collapses space,

• i.e., if it transforms $\vec{i}$ and $\vec{j}$ such that they're colinear.

• Hence, $\det(M) = 0$, i.e., M is non-invertible.

• Because for a matrix to collapse area to zero

$$\text{Hence, for } M\vec{v} = 0 \Rightarrow \det(M) = 0$$

So, setting $\det(M) = 0$

$$\det(A - \lambda I) = 0$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} a-\lambda & b \\ c & d-\lambda \end{bmatrix} = 0$$

$$(a-\lambda)(d-\lambda) - bc = 0$$

$$ad - \lambda a - \lambda d + \lambda^2 - bc = 0$$

$$\lambda^2 - (a+d)\lambda + (ad-bc) = 0$$

$$\lambda = \frac{(a+d) \pm \sqrt{(a+d)^2 - 4(ad-bc)}}{2}$$

$$\lambda = \frac{(a+d) \pm \sqrt{a^2+d^2+2ad-4ad+4bc}}{2}$$

$$\lambda = \lambda_1, \lambda_2 \quad \text{Eigenvalues of matrix } A$$

now for λ_1

$$M\vec{v} = 0$$

$$(A - \lambda_1 I)\vec{v} = 0$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} - \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

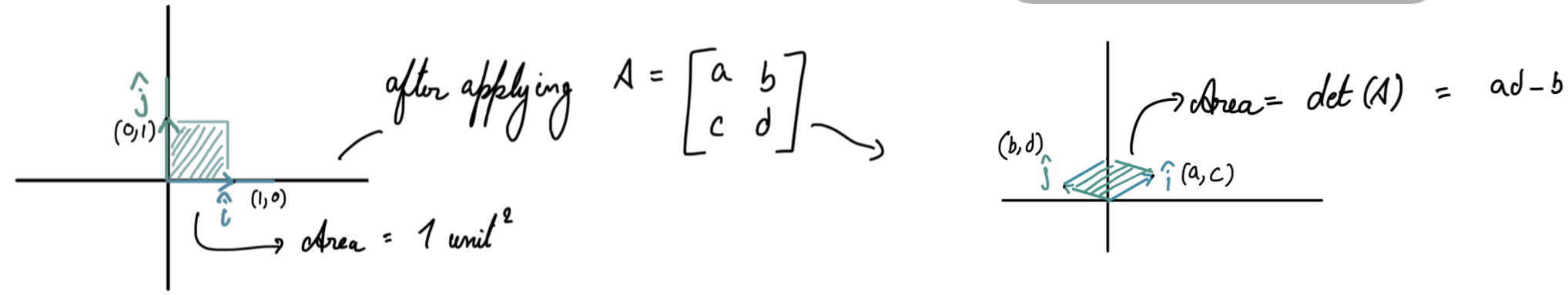
$$\begin{bmatrix} a-\lambda_1 & b \\ c & d-\lambda_1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} (a-\lambda_1)x + by \\ cx + (d-\lambda_1)y \end{bmatrix} = 0 \Rightarrow \begin{aligned} (a-\lambda_1)x + by &= 0 \Rightarrow y = (a-\lambda_1)x/b \\ cx + (d-\lambda_1)y &= 0 \Rightarrow y = cx/(\lambda_1-d) \end{aligned}$$

Determinant

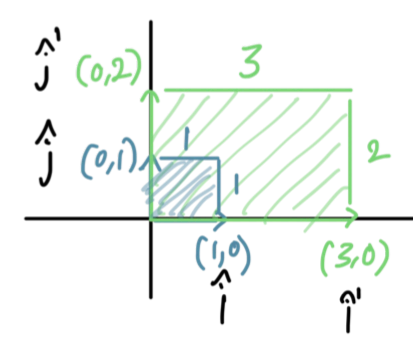
A scalar value computed for a given square matrix.

Geometrically, the determinant of a 2x2 matrix is the
area scaling factor of the transformation.



$$\text{eg } A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \quad \text{new } \vec{i} = \vec{i}' = (3, 0)$$

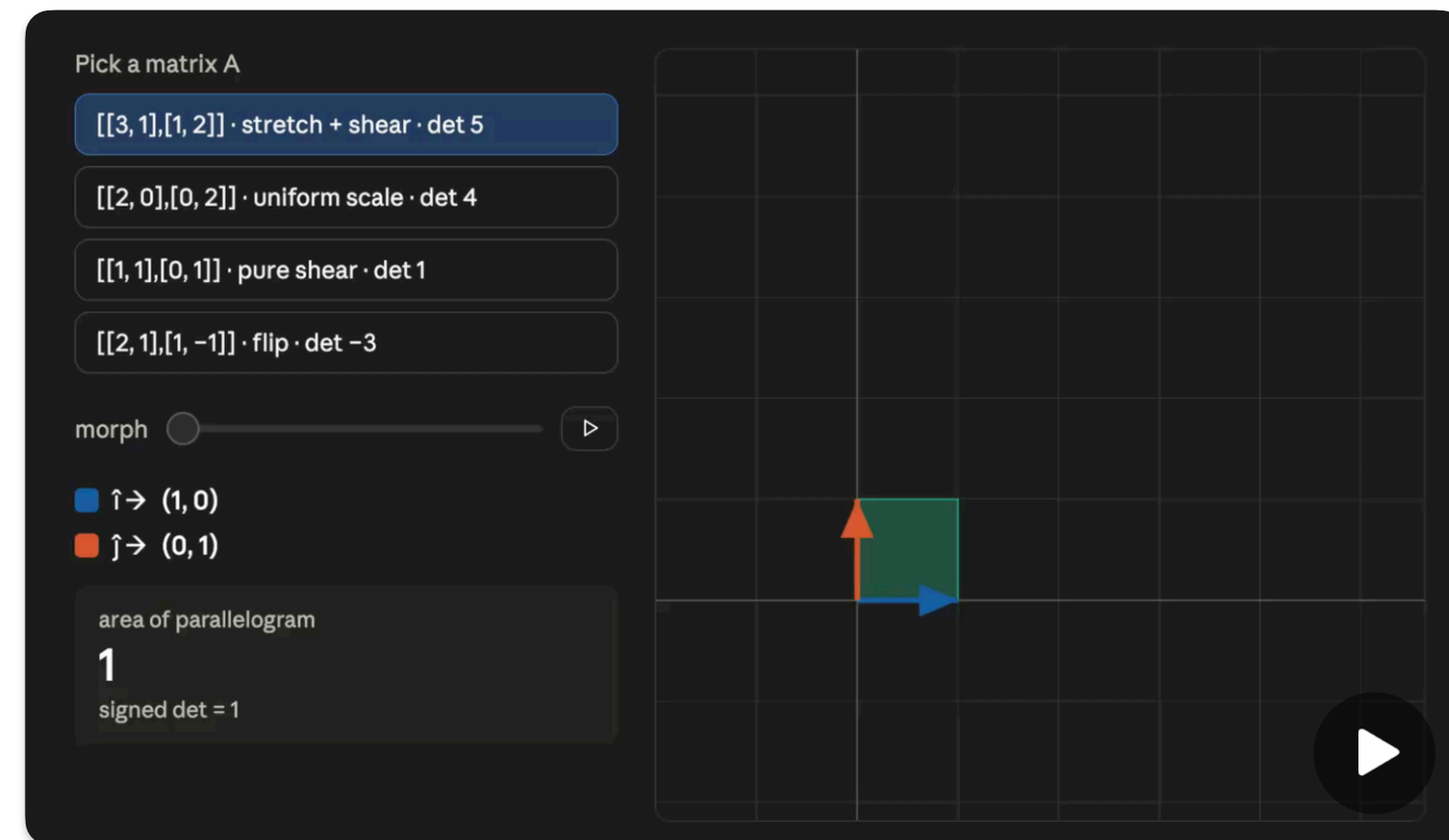
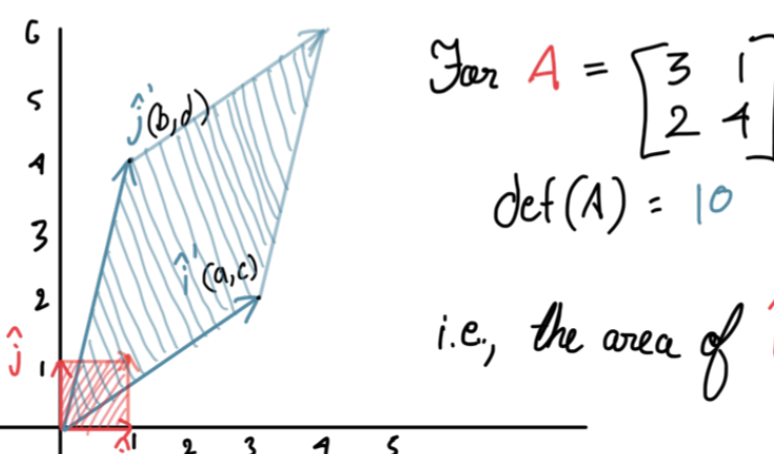
$$\text{new } \vec{j} = \vec{j}' = (0, 2)$$



Area formed by vectors $\vec{i}', \vec{j}' = |\vec{i}' \times \vec{j}'|$

" " $\vec{i}', \vec{j}' = 3 \times 2 = 6$

So, the Determinant of the transformation of
square matrix A —i.e., the factor by which it
scales the area of any region—is 6.



Theorem:

Let $v_1, v_2, \dots, v_n$ be vectors in $\mathbb{R}^n$, let P be the

parallelepiped determined by these vectors, and let

A be the matrix with rows $v_1, v_2, \dots, v_n$.

Then the absolute value of the determinant of A is the volume of P :

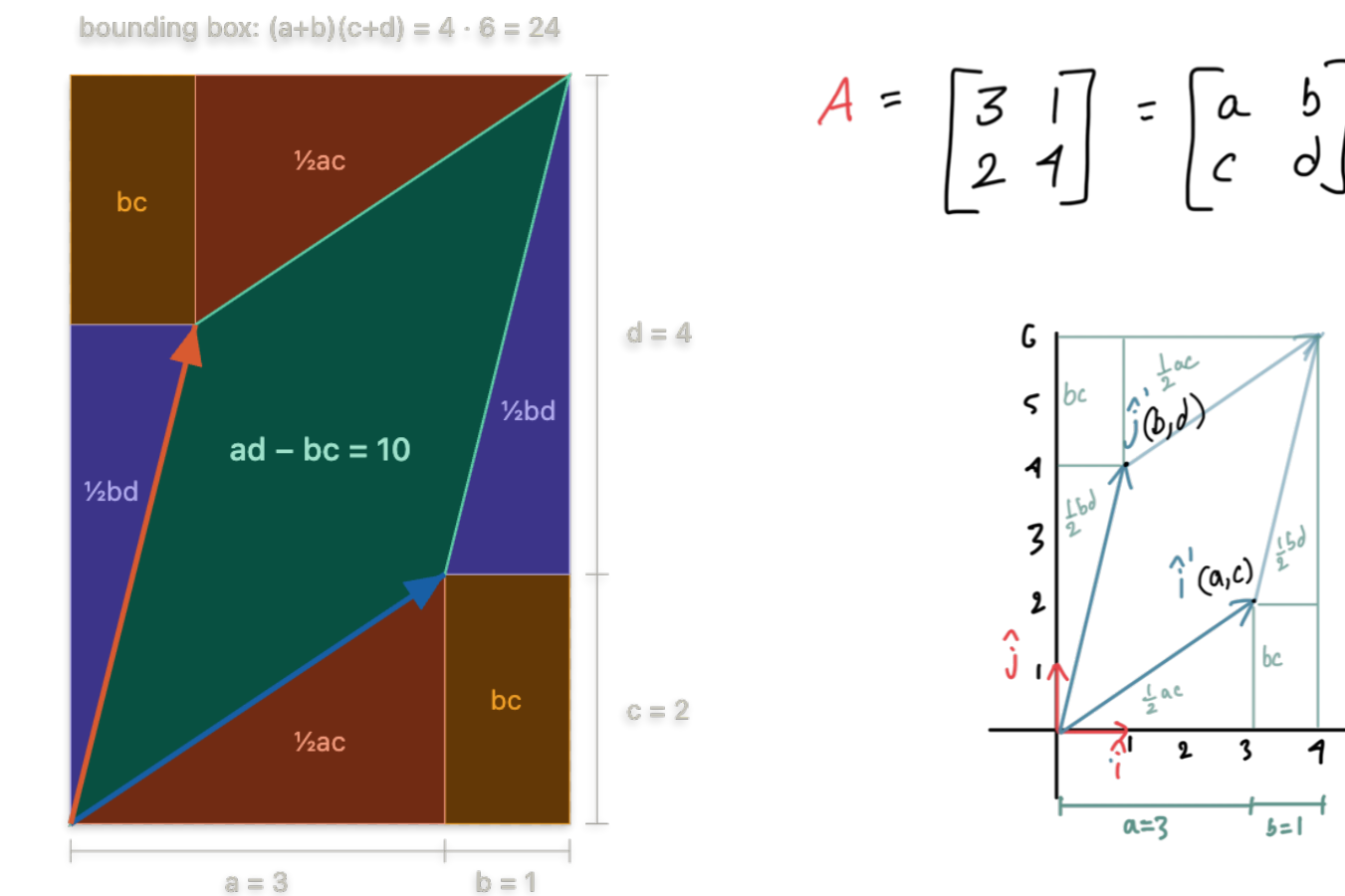
$$|\det(A)| = \text{Vol}(P)$$

Definition

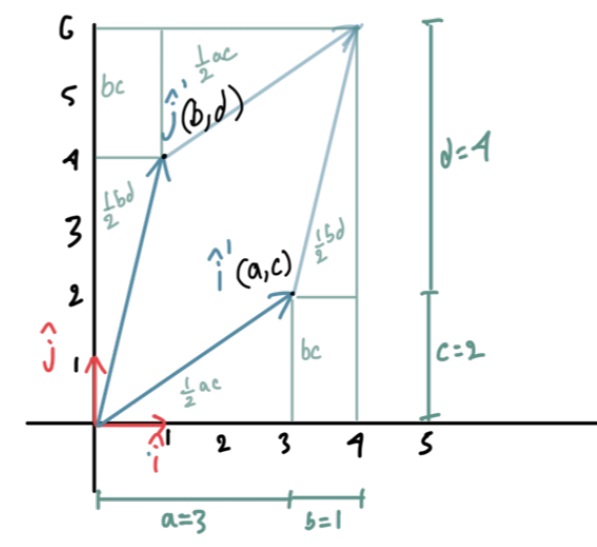
$\det: \{\text{square matrices}\} \rightarrow \mathbb{R}$

Satisfying the following conditions:

- > Scaling a row of A by c multiplies $|\det(A)|$ by $|c|$
- > Doing a row replacement on A does not change A .
- > Swapping two rows of A multiplies $\det(A)$ by -1 .
- > If I is an identity matrix, then $\det(I) = 1$.



$$A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$



Area of Parallelogram:

$$(a+b)(c+d) - \left(2\left(\frac{1}{2}ac\right) + 2\left(\frac{1}{2}bd\right) + 2bc\right)$$

$$ac + ad + bc + bd - (ac + bd + 2bc)$$

the new diagonal
of parallelogram

how much the parallelogram is
stretched or squished along the
new diagonal.